DSA0204
Computer Vision with Open CV

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CO2 - AT3 - Class Test

1. A dark image lacks details. Describe and justify appropriate grey level transformation to enhance visibility, explaining the improve of low-intensity.

Power-Law Transformation:

Transformation increases the brightness of dark pixels while preserving the natural appearance.

Effects:

→ Reveals hidden details.
→ Preserves the natural constrast of brighter areas.
→ Produces a clearer and readable image.

Justification:

Specifically enhances low-intensity pixels without increasing brightness.

Applications:

→ Medical image enhancement
→ Night-time photography
→ CCTV Surveillance

Conclusion:

Power-law Transformation is the most suitable for dark images. It effectively enhances low-intensity regions.

2. A noisy image contains random intensity variations. Explain the suitable smoothing filter.

Gaussian Filter:

Reduces random intensity variations. It smooths the image by giving higher weight to central pixel and gradually lower weights to neighboring pixels.

Effects:

→ Smooths the image uniformly

→ Preserves overall image structure

→ Reduces random intensity noise.

Advantages:

→ Reduces random noise.

→ Produces smooth and natural image

→ Preserves image features.

Conclusion:

Effectively reduces noise using weighted average.

3. An image appears blurred after smoothing. Explain suitable sharpening techniques.

Laplacian filter:

Laplacian filter is a suitable sharpening method because it enhances regions where there is a rapid change in intensity, such as edges and fine details.

→ Enhances edges.
→ Restores fine details
→ Improves image sharpness.

Expected Results:
→ Restores details lost during smoothing
→ Improves edge visibility.
→ Enhances image clarity.

The Laplacian Filter is preferred as it enhances edges by emphasising high-frequency information

4. An image contains high-frequency noise. Explain the appropriate frequency domain filter.

Low-Pass Filter (LPF):

Allows low-frequency components to pass while suppressing high-frequency noise.

Effects:
→ Reduces high-frequency noise
→ Produces a smoother image
→ Preserves the overall image structure.

The LPF is preferred because high-frequency noise mainly exists in high-frequency region of frequency spectrum. By suppressing these components, the filter removes noise while retaining the important image content.

→ Digital photography
→ Remote sensing.

5. A smooth version of an image is required. Describe the frequency domain technique.

Low-Pass Filter (LPF):

LPF is suitable frequency domain technique. It allows low-frequency components to pass while attenuating high-frequency components.

→ Removes high-frequency noise

→ Produces smooth image.

→ Reduces unwanted intensity variations.

After applying LPF:

→ Noise is significantly reduced.

→ Image becomes smoother.

6. An object is clearly brighter than the background. Explain the segmentation technique used.

Global Thresholding:

Global Thresholding is used for object and background have a clear difference in intensity.

Working Process:

Pixel Intensity > T → Object

Pixel Intensity ≤ T → Background

T - Threshold value

Effectiveness:

→ Separates objects from the background.

→ Produces a binary image.

→ Improves object detection.

→ Simplifies further analysis.

→ Simple and fast.

7. Seperate foreground and background. Explain segmentation method.

Thesholding:

→ Pixels with intensity greater than T are classified as Foreground.

→ Pixels with intensity less than T are classified as background.

Jusification:

Threshold is simple, fast and highly effective.

Applications:

→ Object detection

→ Industrial inspection

→ CV systems.

8. An image contains low-frequency background and high-frequency noise.

Low-Pass Filter (LPF):

→ Convert image using Fast Fourier Transform.

→ Spatial domain using Inverse Fast Fourier.

Effects:

→ Removes high-frequency noise.

→ Preserves the image structure.

→ Produces smoother image.

→ Improves image quality.

9. An image requires enhancement of edges without amplifying noise. Explain spatial domain.

Unsharp Masking:

Enhances image edges while controlling the amplification of noise.

Effects:

→ Enhances object edges.

→ Restores fine details.

→ Minimizes excessive noise amplification.

Justification:

Selectively sharpens edges rather than enhancing the entire image.

10. An application requires precise object boundary. Explain segmentation approach.

Thresholding + Canny Edge Detection

Thresholding → Separates object from background

CED → extracts object's boundaries.

Effects:

→ Separates the object accurately.

→ Reduces false object detection.

→ Detects thin edges.

Applications:

→ Shape analysis

→ Object recognition

→ CV Systems.